

Presentation Script

Title: The SUREO Initiative — Protecting the Future of Metro Systems

Good Evening everyone.

Most of us rely on subway systems every day. They quietly keep cities moving. In Tokyo alone, the metro carries more than 6.8 million passengers daily. It is one of the most punctual and efficient systems in the world.

But even systems that appear perfect on the surface can have hidden vulnerabilities underground.

If you live in New York or have taken the New York City subway, you may already understand this idea. Trains can run on time most days, but heavy rain, flooding, or aging tunnels can quickly disrupt service. What passengers experience above ground is reliability, but below ground there is a constant battle against water, corrosion, and aging infrastructure.

This is the challenge my project addresses.

The Problem

Tokyo's metro system is incredibly efficient today. However, climate change and infrastructure aging are creating new risks.

More intense rainfall increases the likelihood of flooding in underground tunnels, and older infrastructure becomes more vulnerable to corrosion and structural stress.

A helpful way to think about this is the New York subway during major storms. When water enters tunnels or stations, trains slow down or stop entirely. Repairs are expensive, and service disruptions affect millions of commuters.

Tokyo faces similar risks. The difference is that the city has an opportunity to act before these problems become emergencies.

The SUREO Strategy

The proposal I developed is called SUREO, which stands for Smart Urban Resilience and Energy Optimization.

The strategy focuses on four main improvements.

First, AI-based flood detection systems. These sensors monitor pressure, humidity, and water levels in real time so that flooding risks can be detected early.

Second, expanded ventilation systems that improve airflow inside tunnels and stations. This reduces humidity and corrosion while also lowering energy use.

Third, predictive maintenance sensors that monitor the condition of infrastructure and identify corrosion or stress before failures occur.

Finally, energy regeneration systems, which capture energy from braking trains and reuse it within the system, making the metro more efficient and environmentally friendly.

Instead of reacting to emergencies, this approach allows operators to prevent problems before they happen.

Implementation and Investment

The proposal outlines a five-year phased implementation plan.

The first year focuses on mapping high-risk areas and testing sensor systems in the most vulnerable stations.

Years two and three expand the deployment of sensors and ventilation upgrades across the network.

The final two years integrate all systems together so the metro can monitor infrastructure conditions in real time.

The total investment is approximately ¥18 billion, or about \$120 million.

While that sounds large, the long-term savings are significant. Preventative monitoring could avoid up to ¥10 billion in emergency repair costs and reduce energy consumption by about 15 percent.

The Impact

Over five years, the project aims to reduce flood risk exposure by 40 percent, reduce energy usage, and improve response times to infrastructure issues.

The key idea is simple:

Cities should not wait for infrastructure to fail before fixing it.

Just like the New York subway continues investing in upgrades after storms like Hurricane Sandy, Tokyo can strengthen its system today to avoid future disruptions.

Closing

Subway systems are the lifelines of modern cities.

By combining real-time sensors, predictive maintenance, and energy-efficient technology, the SUREO initiative shows how transit systems can become more resilient, more sustainable, and more reliable for the millions of people who depend on them every day.

Thank you.

Reflection Paper

Working on this presentation required me to rethink how I explain technical ideas to a general audience. My original research focused on infrastructure resilience in Tokyo's metro system, which included technical concepts such as predictive maintenance sensors, climate risk modeling, and infrastructure corrosion monitoring. While these ideas make sense in engineering or policy discussions, they can be difficult for a general audience to immediately understand.

I used AI tools to help simplify complex explanations and restructure the content into clearer narratives. One of the most helpful aspects was using AI to identify which technical details could be removed or simplified without losing the core message. For example, instead of describing detailed sensor architectures or engineering specifications, the presentation focuses on the idea of "detecting problems early before they become emergencies." That framing makes the technology easier to grasp.

Another important adjustment was adding comparisons to the New York City subway. Since many people in the audience are likely familiar with subway disruptions during storms or infrastructure repairs, the NYC example helps translate Tokyo's situation into something relatable. This comparison made the presentation feel less abstract and helped anchor the discussion in a real-world experience people recognize.

The visuals in my slides also played an important role in supporting the explanation. Rather than relying on dense text, the slides use diagrams, images of tunnels, and simplified charts. These visuals help show the difference between surface reliability and hidden underground risks. They also illustrate the four components of the SUREO strategy in a way that viewers can quickly understand. I designed the slides so that they reinforce the main idea while the spoken script provides the detailed explanation.

I also revised the presentation based on feedback and AI suggestions to improve clarity and pacing. Early drafts were too technical and sounded more like a research report than a presentation. After revising, the script now focuses more on storytelling, beginning with the everyday experience of subway riders and then gradually introducing the problem and solution.

Overall, AI was helpful as a brainstorming and editing tool, but I made sure to adjust the tone and wording so the final presentation felt natural and clear. My goal was to create something that a non-technical audience could easily follow while still communicating the importance of infrastructure resilience.